

INTRODUCTION

Blood transfusions save around 118.5 million lives a year worldwide. The availability of O blood impacts this statistic. The world population blood supply is split up into four groups; A, B, AB, and O. 35.2%, 16%, 1% (rarest), and 7% respectively. For blood transfusions, O negative is the most useful due to its lack of an antigen layer. An antigen layer is unique to each blood group as they detect foreign substances in the blood stream.

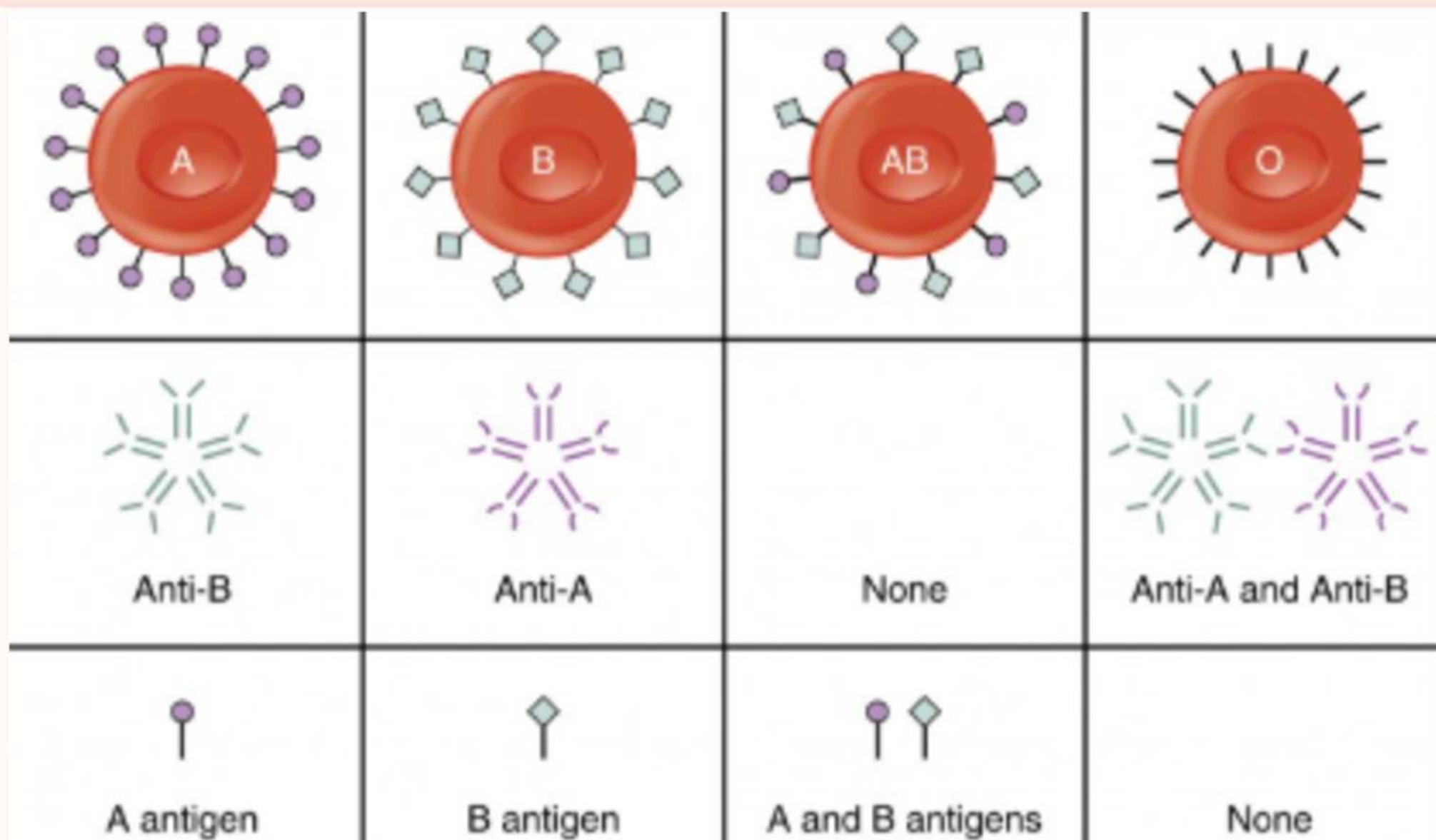


Figure 1: showing different blood characteristics
Bigod Typing | Anatomy and Physiology II. (n.d.). Lumen Learning.
Retrieved March 4, 2026, from
<https://courses.lumenlearning.com/suny-ap2/chapter/blood-typing/>

BACKGROUND

O negative is used in emergencies due to limited time when a patient is in critical condition. Since, O negative is the universal donor and is compatible with all other blood types, no prior test is needed. Unfortunately there is a shortage of O blood and donors.

PURPOSE

The purpose of this research was to investigate a procedure that can potentially transform any blood type into O negative.

TRANSFORMATION OF GROUP B BLOOD TO GROUP O THROUGH PLASMA TRANSFERS AND ENZYMATIC PROCESSES

METHOD

PROCEDURE ONE:

B plasma sheep's blood was centrifuged and B negative blood was added to it so the final mix contained both A and B plasma. No separation in the vials was noted indicating that this part of the procedure was successful.



Figure 2: Image taken by researcher showing the transfer of plasma to blood

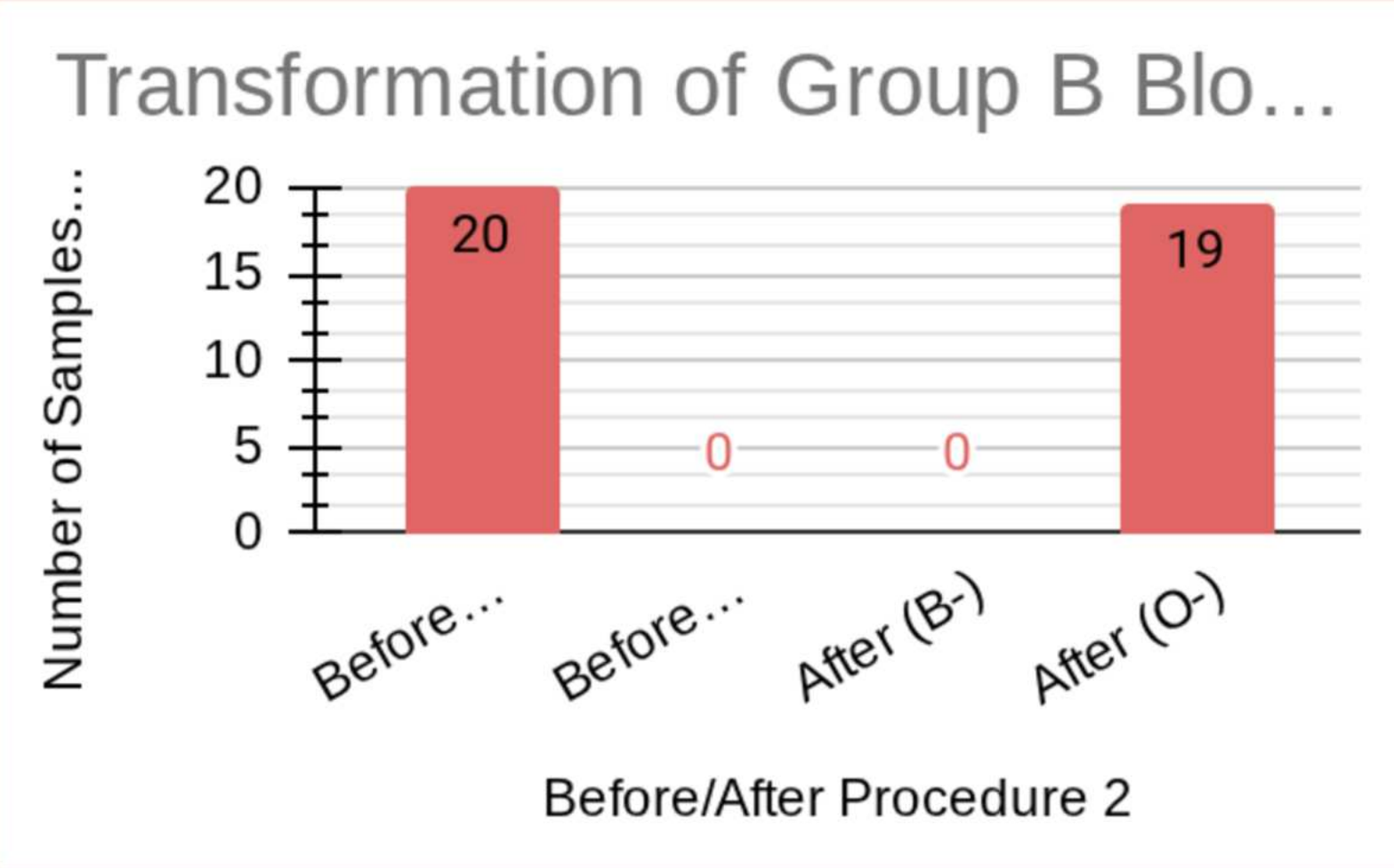


Figure 3: Image made by researcher from Google Spread Sheet showing the data of Procedure One



Figure 4: Image taken by researcher showing completion of Procedure One

PROCEDURE TWO:

B plasma sheep's blood was centrifuged and B negative blood was added to it so the final mix contained both A and B plasma. No separation in the vials was noted indicating that this part of the procedure was successful. Data indicated that 19 of the 20 samples were O negative, indicating 95% successful transformation.

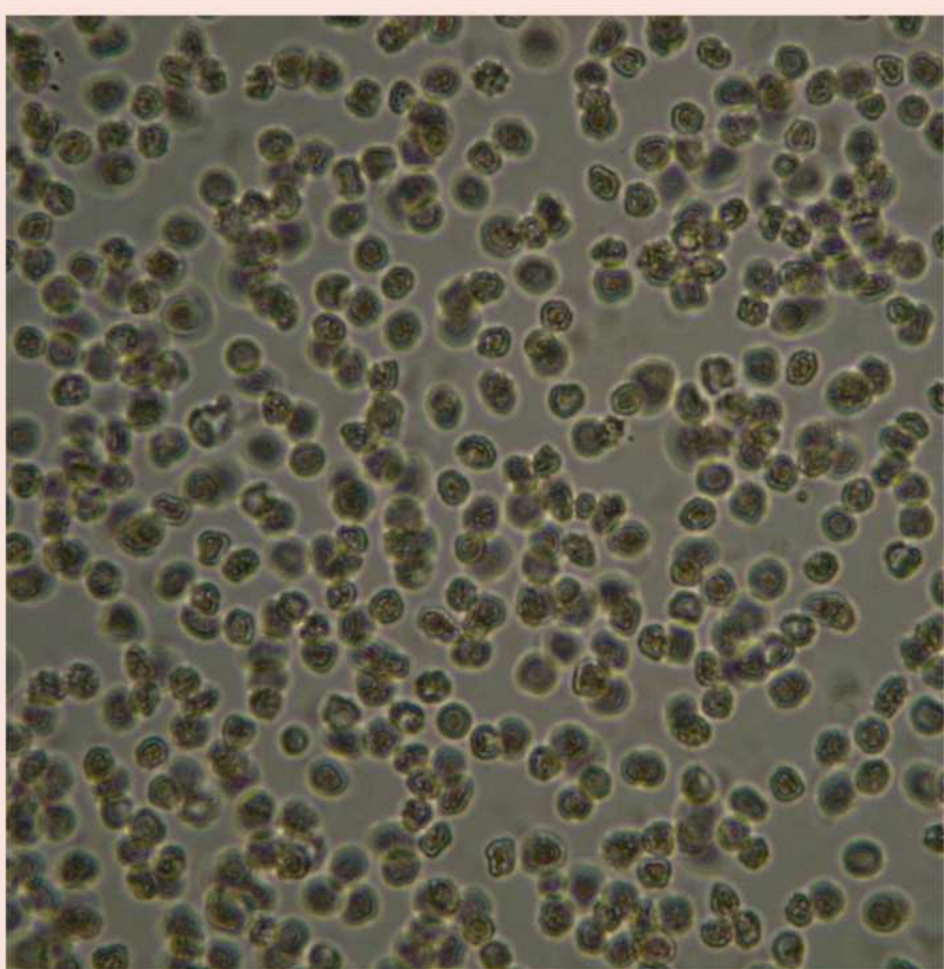


Figure 5: Image taken by researcher showing blood cells still intact after Procedure One

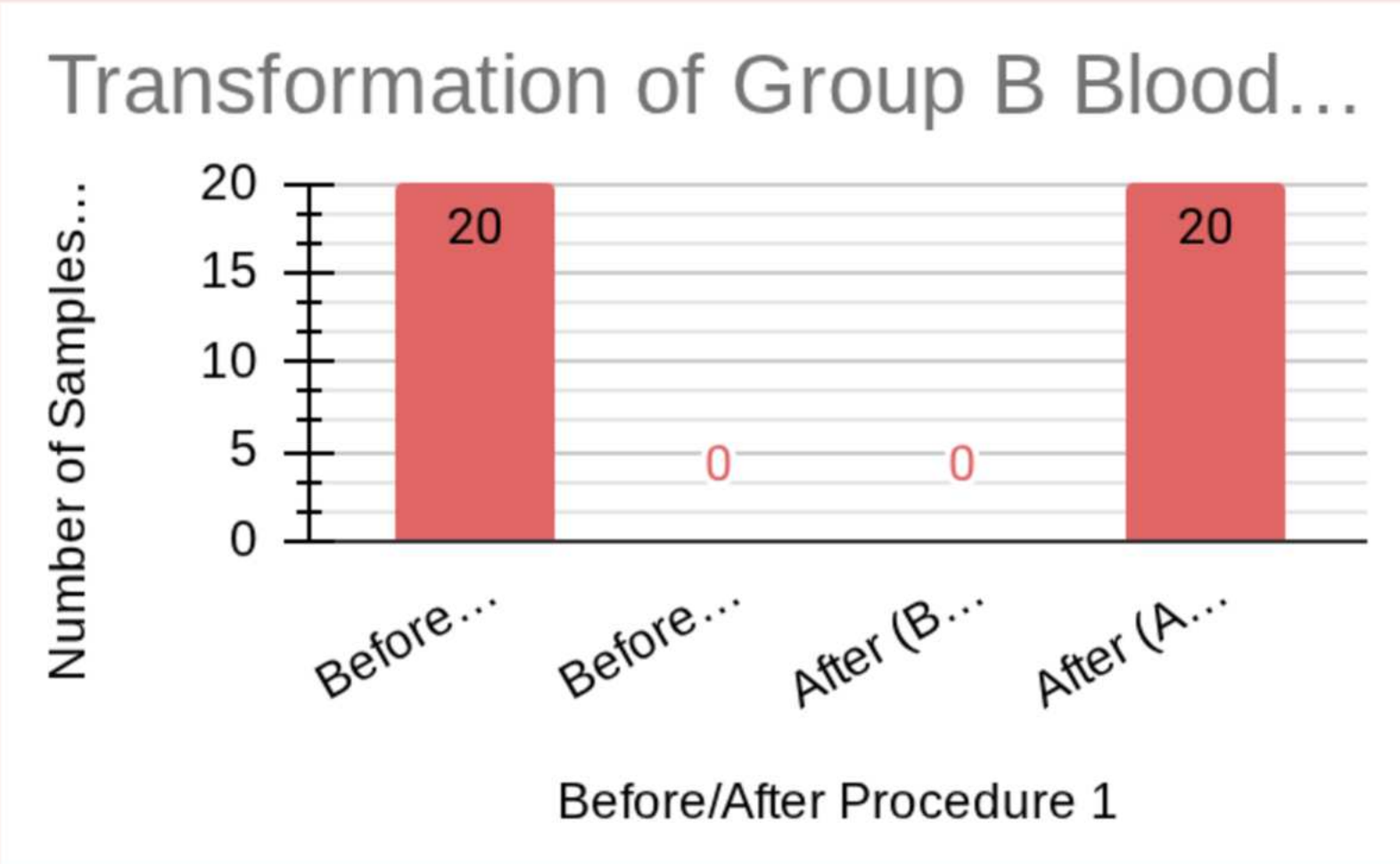


Figure 6: Image made by researcher from Google Spread Sheet showing the data of Procedure Two

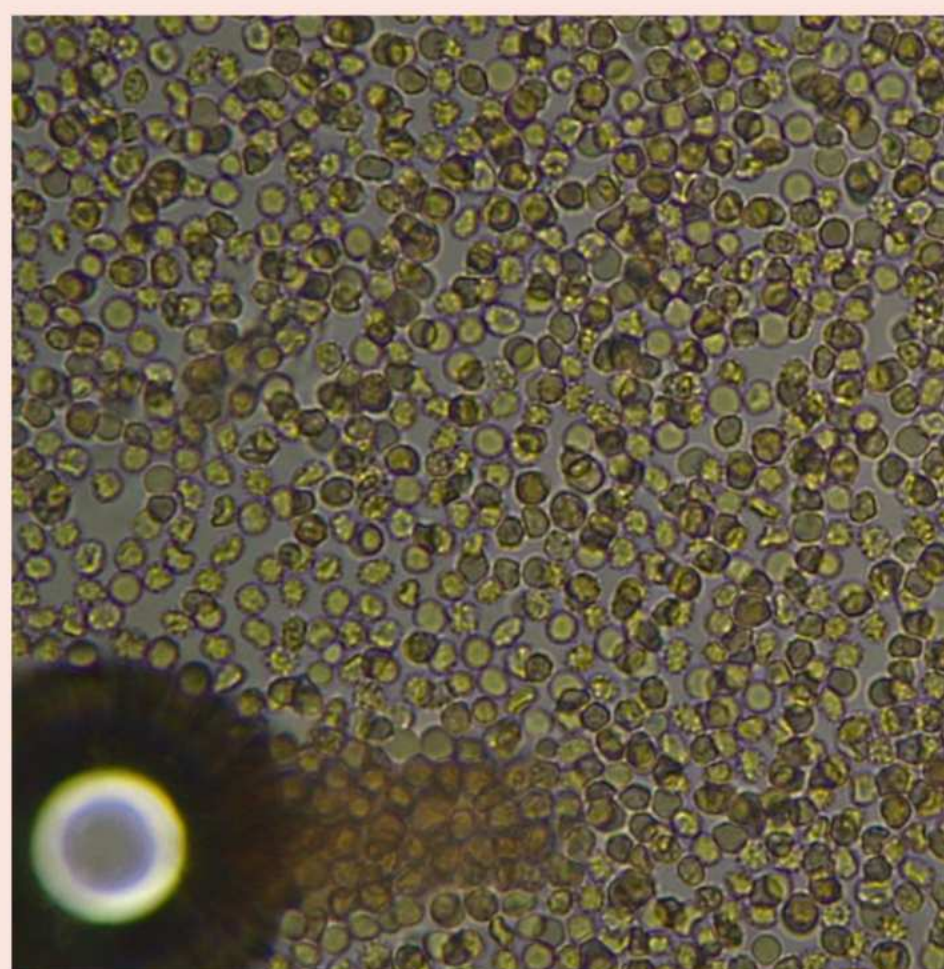


Figure 7: Image taken by researcher showing healthy blood cells after Procedure Two

CONCLUSION

This experiment may help pave the way for ongoing research to be continued so that O negative blood becomes readily available around the world.

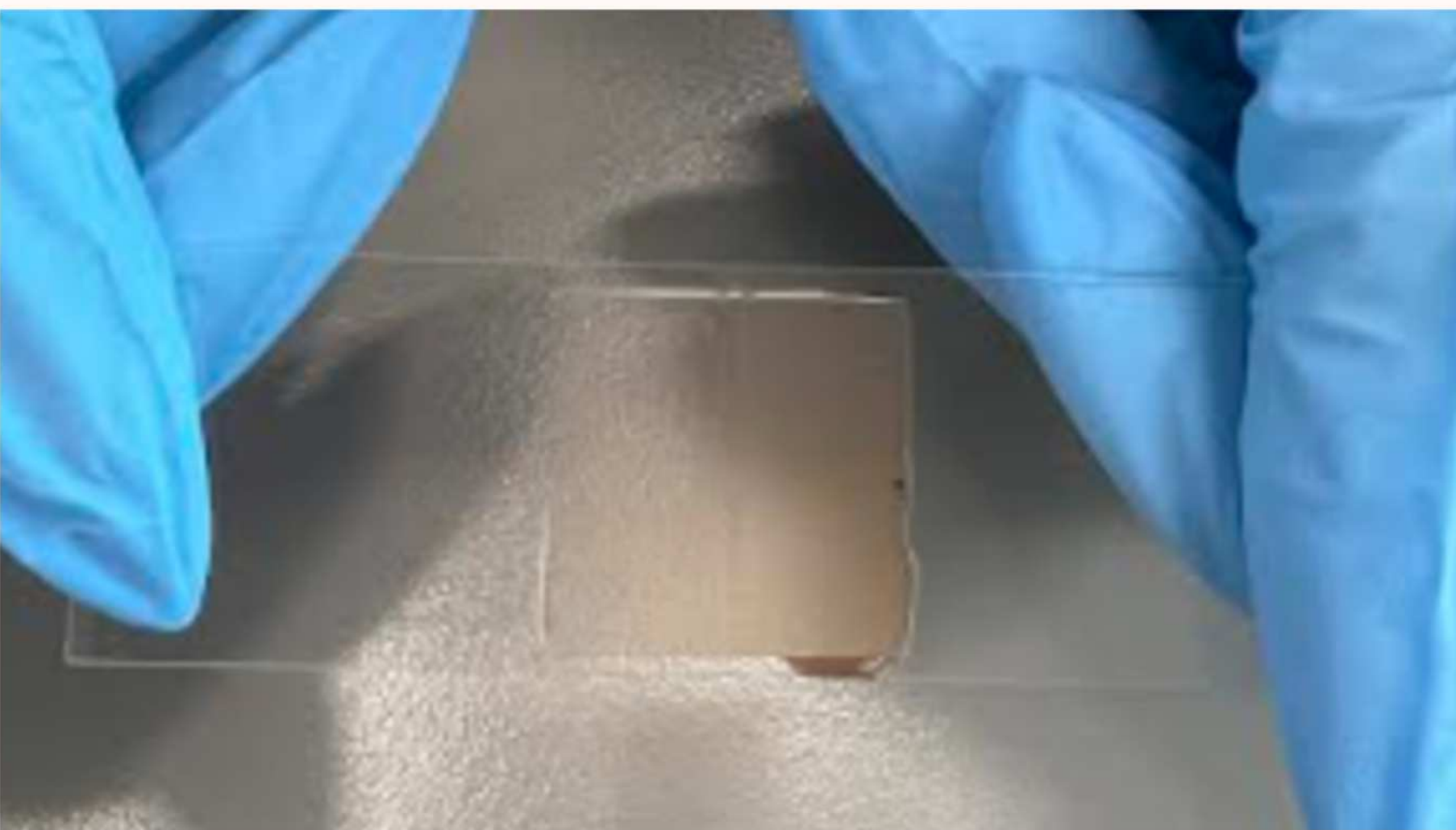


Figure 8: Image taken by researcher showing blood sample slide



Figure 9: Image taken from researcher crushing alpha- galactosidase enzymes into powder form



Figure 10: Image taken by researcher extracting plasma and blood to add enzymes

REFERENCES

Blood Plasma and Serum Preparation | Thermo Fisher Scientific – US (n.d.). ThermoFisher. Retrieved March 2, 2026, from <https://www.thermofisher.com/us/en/home/references/protocol/s/cell-and-tissue-analysis/elisa-protocol/elisa-sample-preparation-protocols/blood-plasma-serum-preparation.html>
Blood Types Explained – A, B, AB and O | Red Cross Blood Services. (n.d.). Red Cross Blood Donation. Retrieved March 2, 2026, from <https://www.redcrossblood.org/donate-blood/blood-types.html>
Pennisi, E. (2019, June 10). Type A blood converted to universal donor blood with help from bacterial enzymes. science.org. Retrieved November 14, 2026, from Type A blood converted to universal donor blood with help from bacterial enzymes